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Modul 5:

Erweiterte Gleichstromnetzwerke

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1 Introduction

While the calculations for series connections without parallel-connected components are relatively straightforward, the analysis of direct current networks with nodes that have n branches, where the number of branches is $n \geq 2$, quickly becomes confusing as the number of nodes increases. A multi-layered consideration of the DC network is necessary here. The following methods are available for calculating DC network:

- Node and mesh rule
- Superposition
- Node potential analysis
- Mesh current analysis

The analysis of electrical networks with more than one current or voltage source is also not trivial. Determining all currents and voltages in the electrical network shown in Figure 1.1 is not possible by calculating series and parallel connections alone.

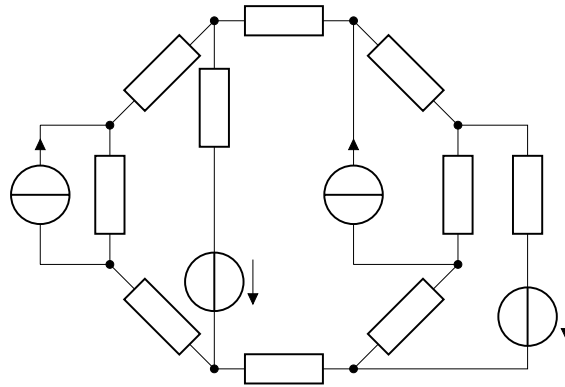


Figure 1.1: **Example Network.** Direct current electrical network with a plurality of resistors, current sources and voltage sources.

2 Knot- and Mesh analysis

When considering electrical networks, the currents in nodes and the voltages in meshes are analysed in particular. The analysis of nodes and meshes in electrical networks is carried out using the two Kirchhoff's rules.

Lernziele: Knot- and Mesh analysis

The Students

1. know the basic definitions for describing an electrical network.
2. can apply the node rule and the mesh rule to electrical networks.

2.1 Current density in free space

Facts and concepts from field theory lead to explanations in structured field space. To this end, Figure 2.1 shows a positively charged plate and a negatively charged plate facing each other. The space between the two plates has an electric field $\vec{E}$ and a medium with conductivity σ . Charge carriers can move in the free space. A directed movement of charge carriers and thus a current density $\vec{J}$ between the plates occurs. The current density is source-free in the area under consideration.

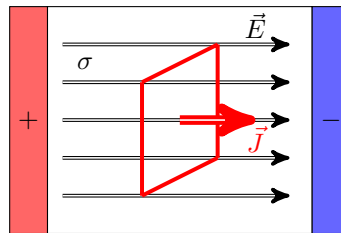


Figure 2.1: **Electric field between a positively charged and a negatively charged plate.** The current density of a sample area in the electric field is source-free.

The source-free nature of the current density is illustrated by the equation 2.1. This describes that no sources or sinks exist, as there are no field-forming charges in space. In this case, the current density is source-free and the same number of charges flow into and out of the surface. Mathematically, this is expressed in the continuity equation for consideration in free space:

$$\oint \vec{J} d\vec{A} = 0 \quad \rightarrow \quad \text{div} \vec{J} = 0 \quad (2.1)$$

2.2 Current density in structured space

In a structured field space (see Figure 2.2), the current density must also be free of sources. All charges that enter the field space must also leave it again. This also applies in cases where an electrical conductor has more than one discharge path. A supply line with two conductors is indicated, in which the charges are distributed between the two conductors. The cylinder surface areas A_1 , A_2 and A_3 define a structured field space with the adjacent cylinders. Just as the charges are distributed,

the current densities are distributed throughout the structured space. Thus, the current density $\vec{J}_1$ is the sum of the outgoing current densities $\vec{J}_2$ and $\vec{J}_3$.

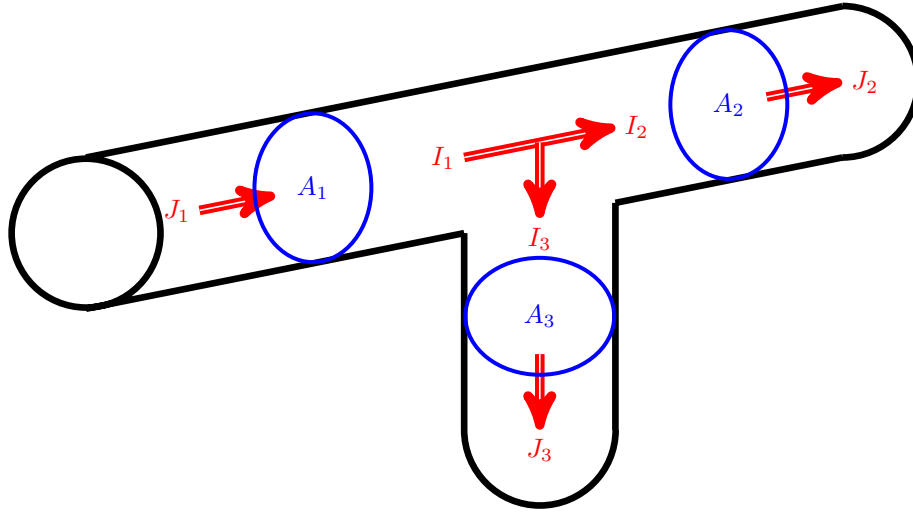


Figure 2.2: **Structured field space.** Simplified representation of a splitting conductor with current densities J_1 , J_2 and J_3 and areas A_1 , A_2 and A_3 .

Since the field space in the example is structured by the surfaces and the cross-section of the conductor is defined, the currents of the conductors can be determined from the relationship between the current density and the surface area according to equation 2.2.

$$I_1 = \iint_A \vec{J}_1 d\vec{A} \quad (2.2)$$

The relationship between the current densities can thus be transferred to the currents. The current of the supply line I_1 is the sum of the two branch currents I_2 and I_3 . The example conductor can also be represented as a simplified electrical network. The currents determined around the connection point K (node) are shown in Figure ??.

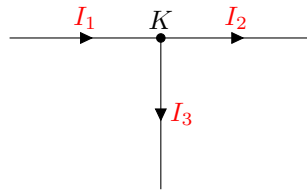


Figure 2.3: **Example nodes from the structured space.** The current densities J_1 , J_2 and J_3 result in the currents I_1 , I_2 and I_3 around node K.

2.3 Digression: Graph theory

In mathematics, graph theory deals with the description of nodes and edges, while account analysis also defines nodes and branches. Graph theory allows the edges and nodes shown in Figure 2.4 and their properties to be related to each other. Multiple nodes can be connected to each other via the edges.

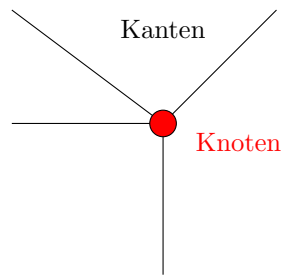


Figure 2.4: **Example nodes in graph theory.** Node with 4 edges to explain nodes and edges in graph theory.

When several nodes are connected to each other with edges, a graph is created. This graph can be used to describe an electrical network, for example. The example of a network with five nodes and the edges between the nodes is shown in Figure 2.5. When searching for a path through the network, there are various possibilities. A path from node 1 via nodes 2, 4 and 5 back to node 1 is highlighted in red. If the start and end are the same node, this path is also referred to as a cycle in graph theory. Any cycle in which each node is "visited only once, except for the start and end, is called a "circle". In electrical networks, these circles are also referred to as meshes.

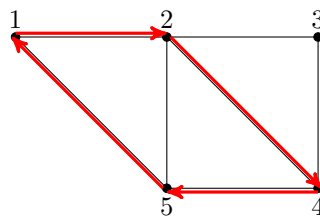


Figure 2.5: **Example network from graph theory.** Network with 5 nodes and edges running between the nodes to explain a cycle in graph theory.

In mathematics, a matrix is a useful tool for explaining the relationship between nodes. The nodes have a differentiated number of edges, and not all nodes are connected to each other. The adjacency matrix describes the relationship between the nodes. Such an adjacency matrix is set up in equation 2.3 for the example network described. It thus completely describes the graph, apart from the graphical representation in the 2D display.

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \end{bmatrix} \quad (2.3)$$

Merke: Graphentheorie

In graph theory, networks are described using edges and nodes. When analysing networks, paths and cycles are defined. A continuous cycle with an identical starting point and end point is referred to as a circle.

2.4 Knot, Branch, Mesh

An electrical network consists of nodes, branches and meshes. A branch connects exactly two nodes by means of one or more circuit elements. The same current flows through all elements of a branch. There are three branches in Figure 2.6. One branch with the components R_1 , R_5 and U_{q1} , one branch with the resistance R_4 and one branch with the components R_2 , R_3 and U_{q2} . A node is a point at which at least three connections of the circuit elements converge. The current can therefore split here. The electrical potential is identical for all connected terminals.

In the circuit diagram, a node is indicated by a filled circle. However, if two or more of these circles are connected to each other by only one line, it is a single node, since the potential is also the same here. A node is denoted by K_n .

A mesh is a closed path consisting of at least two branches. In the net in Figure 2.6, three meshes can be defined:

- M_1 consisting of R_1 , R_4 , R_5 and U_{q1}
- M_2 consisting of R_2 , R_3 , R_4 and U_{q2}
- M_3 consisting of R_1 , R_2 , R_3 , U_{q2} , R_5 und U_{q1}

A mesh is designated by M_n . The direction of rotation is important and is indicated by an arrow.

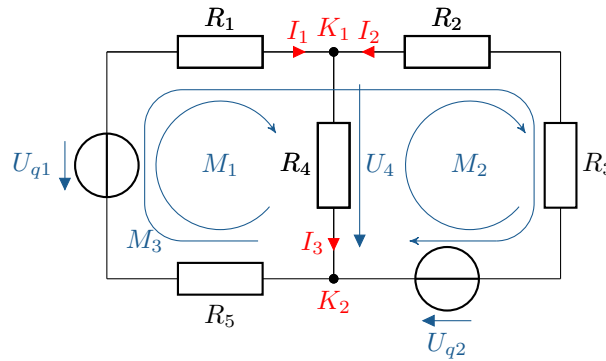


Figure 2.6: **Direct current network.** Network with **Knots**, Branches and **Meshes**



Merke: Knots, Branches and Meshes

Electrical networks are described by nodes, branches and meshes, analogous to graph theory.

2.5 The complete tree

For further analysis of the network, it is necessary to determine the network equations. However, this linear system of equations is always overdetermined, which is why it is necessary to reduce the number of equations. It is important to identify and eliminate the linearly dependent equations.

The nodes, branches and meshes of a network can be represented in a graph that only shows the connections between them. To represent the graph, all branches are shown as lines that connect the nodes. The content of the branch is irrelevant for this purpose. Figure 2.7 shows an example network with the resulting graph.

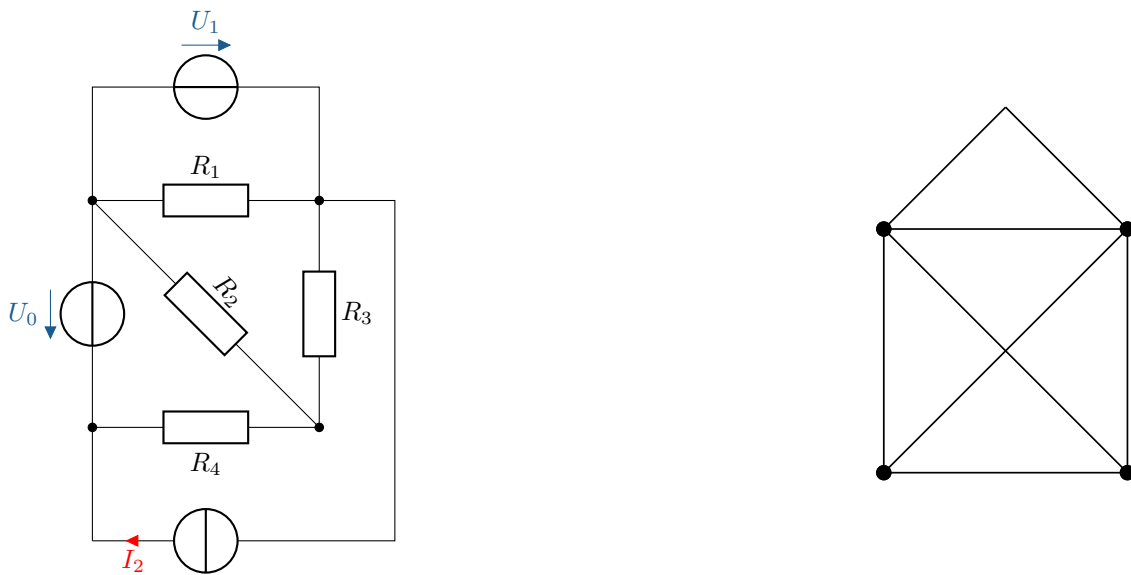


Figure 2.7: **Direct current network and graph.** Generation of the graph from a network to be calculated

In this graph, a complete tree can be drawn that contains all nodes but does not itself form a mesh. The tree branches connect all nodes to each other but do not form a closed line. The tree branches can form branches, meaning that a node can also touch more than two tree branches. In larger networks, there are several possibilities for a complete tree, which are in principle equivalent. However, when presenting the various calculation methods, certain combinations are preferred depending on the method. A complete tree contains exactly $k - 1$ branches.

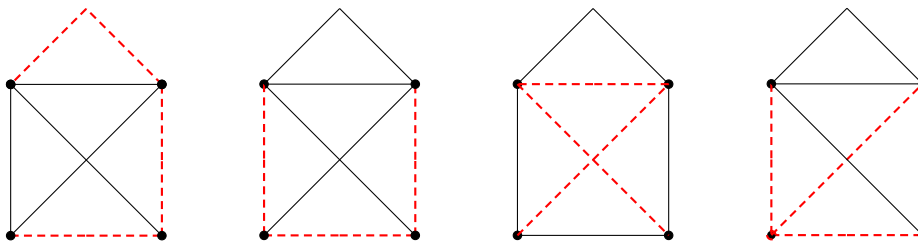


Figure 2.8: **Example trees.** Examples of different complete trees of a network drawn in red

The branches belonging to the tree are called tree branches (red), the other connecting branches (black). To find the linearly independent meshes, a mesh is formed for each connecting branch that contains only tree branches except for the connecting branch.

- A network with z branches and k nodes contains $z - k + 1$ connecting branches.



Merke: The complete tree

With the help of a complete tree, all nodes are connected to each other by tree branches. This always results in $k - 1$ tree branches. The complete tree does not form a mesh.

2.6 Knot rule

If more than two lines meet at a point in an electrical network, this is referred to as a node K. The direction of the arrow from the perspective of the node determines the sign of the current. If a current flows into a node, the current arrow points towards the node and this is referred to as an inflow. If the current arrow points out of the node, the current flows out of the node and this is an outflow. The node analysis is based on Kirchhoff's first law, which states: The sum of all currents at a node is zero. (Gleichung 2.4). This law follows directly from the continuity equation. (vgl. Abschnitt [Current density in free space](#))

$$\sum_{k=1}^N I_k = 0 \quad (2.4)$$

Thus, for each node, the sum of the currents from inflows and outflows must balance each other out. Figure 2.9 shows a node K and the inflows and outflows. The flows I_1 and I_2 are inflows. The three flows I_3 , I_4 and I_5 are outflows.

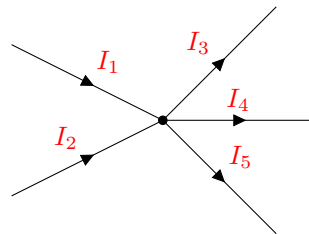


Figure 2.9: **Example node.** Node for node analysis with two incoming and three outgoing branches.

If the currents are applied one after the other and set to zero, the equation 2.5 is obtained. Depending on whether it is an inflow or an outflow, the sign is selected. Inflows are assigned a positive sign and outflows a negative sign.

$$\sum_{k=1}^N I_k = 0 = I_1 + I_2 - I_3 - I_4 - I_5 \quad (2.5)$$

Merke: Knot rule

The sum of the incoming flows is equal to the sum of the outgoing flows, or the sum of all flows at a node is zero:

$$\sum_{k=1}^N I_k = 0 \quad (2.6)$$

2.7 Mesh rule

When a particle moves between two points in an electric field, work is done. If the starting point is identical to the end point, the same amount of work must be delivered as absorbed. In this case, the sum of the work is zero. If such a complete mesh cycle is performed along a closed path, the sum of the partial voltages must also be zero. This assumes that there is no time-varying magnetic flux

density and therefore no electrical eddy field is created (see equation 2.7). The equation describes Faraday's law of induction for static fields (here: no change in a magnetic field).

$$\oint_s \vec{E} d\vec{s} = 0 \quad \rightarrow \quad \text{rot} \vec{E} = 0 \quad (2.7)$$

According to Kirchhoff's second law, mesh analysis states that for a complete circuit (mesh) in an electrical network, the sum of all voltages is equal to zero. This is illustrated by the equation 2.8.

$$\sum_{k=1}^N U_k = 0 \quad (2.8)$$

Within a mesh, voltages with an arrow direction in the same direction are counted as positive, while voltages with an arrow direction opposite to the direction of rotation are counted as negative. Based on the current curves and potentials from Figure 2.10, the voltages of the respective branches can be determined. There is no potential difference above the parallel connection of the resistors. The same applies to the electrical connection below the parallel connection. The current I_{R1} flows through the resistor R_1 . According to Ohm's law, this results in the voltage U_{R1} across R_1 . Equivalently, the voltage U_{R2} across R_2 is determined.

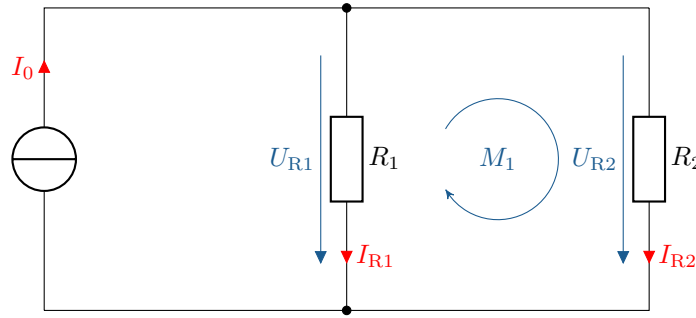


Figure 2.10: **Example network with one power source and two parallel resistors.** In mesh M_1 , the voltages U_{R1} and U_{R2} must be identical and cancel each other out according to their directions in the mesh.

According to the network presented and the direction of the drawn mesh M_1 , the relationship according to equation 2.9 applies. The voltage U_{R2} runs in the same direction as the drawn mesh and is assigned a positive sign. The voltage U_{R1} runs in the opposite direction to the mesh M_1 and is therefore negative. In total, these two voltages must add up to zero.

$$\sum_{k=1}^N U_k = +U_{R2} - U_{R1} = 0 \quad (2.9)$$

Merke: Mesh rule

In a mesh, the sum of all partial stresses is equal to zero.:

$$\sum_{k=1}^N U_k = 0 \quad (2.10)$$

2.8 Knot and Mesh analysis

In node and mesh analysis, the complete system of equations consisting of node and mesh equations is set up and then solved. The procedure is simple, but generates a number of equations for a network that corresponds to the number of branches. The procedure is as follows:

1. Simplifying the network.
2. Draw all current arrows and create the graph.
3. Set up $k - 1$ node equations. Any node equation can be omitted, since in a network with k nodes only $k - 1$ node equations are linearly independent of each other. In principle, it does not matter which node equation is omitted. It is best to omit the most complicated equation.
4. Set up the $m = z - k + 1$ linearly independent mesh equations. The equations are created using the complete tree.
5. Solving the system of equations with the dimension z .
6. If necessary, undo step 1.

The following example network is considered:

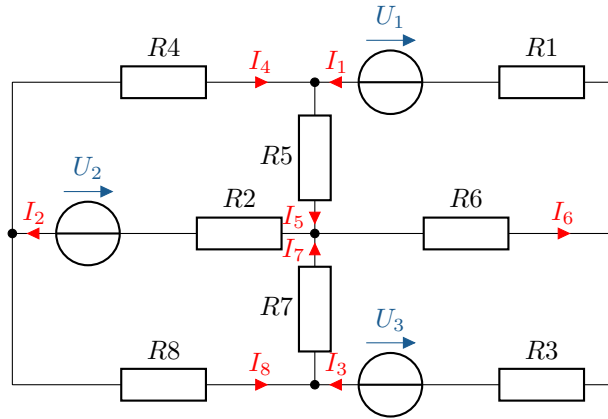


Figure 2.11: **Example network for node and mesh analysis.**

The corresponding graph and the meshes defined by it are shown in Figure 2.12. A different arrangement of the meshes is possible. As shown, meshes can also extend beyond branches.

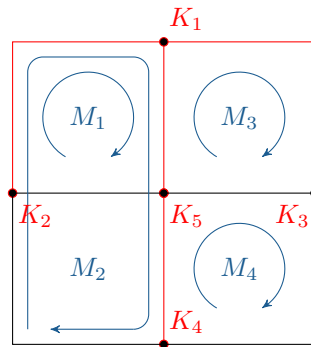


Figure 2.12: **Graph with mesh, for example.** The tree branches are coloured red and the mesh branches are coloured blue.

The network has $z = 8$ branches and $k = 5$ nodes. Therefore, $k-1 = 4$ node equations and $z-k+1 = 4$ mesh equations must be set up. Node 5 has four branches, the others only three. Therefore, the node equation for node 5 is omitted.

$$\begin{aligned}
 K_1 : \quad & I_1 + I_4 - I_5 = 0 \\
 K_2 : \quad & I_2 - I_4 - I_8 = 0 \\
 K_3 : \quad & -I_1 - I_3 + I_6 = 0 \\
 K_4 : \quad & I_3 - I_7 + I_8 = 0 \\
 M_1 : \quad & R_2 I_2 + R_4 I_4 + R_5 I_5 - U_2 = 0 \\
 M_2 : \quad & R_4 I_4 + R_5 I_5 - R_7 I_7 - R_8 I_8 = 0 \\
 M_3 : \quad & -R_1 I_1 - R_5 I_5 - R_6 I_6 + U_1 = 0 \\
 M_4 : \quad & R_3 I_3 + R_6 I_6 + R_7 I_7 - U_3 = 0
 \end{aligned}$$

To make the system of equations easier to solve and to increase clarity, it can be represented in matrix notation. With a little practice, matrix notation can also be used immediately.

$$\begin{bmatrix}
 1 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\
 0 & 1 & 0 & -1 & 0 & 0 & 0 & -1 \\
 -1 & 0 & -1 & 0 & 0 & 1 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 & 0 & -1 & 1 \\
 0 & R_2 & 0 & R_4 & R_5 & 0 & 0 & 0 \\
 0 & 0 & 0 & R_4 & R_5 & 0 & -R_7 & -R_8 \\
 -R_1 & 0 & 0 & 0 & -R_5 & -R_6 & 0 & 0 \\
 0 & 0 & R_3 & 0 & 0 & R_6 & R_7 & 0
 \end{bmatrix} \cdot \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \\ I_7 \\ I_8 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ U_2 \\ 0 \\ -U_1 \\ U_3 \end{pmatrix}$$

This 8th order system of equations can now in principle be solved using the methods known from mathematics. However, we do not want to do that here, as simpler methods are presented below that are easier to calculate.

Example 2.1: Network analysis

The electrical network shown in Figure 2.13 is given. The following tasks are to be completed:

- Plotting currents and voltages
- Setting up the node equations
- Setting up the mesh equation M_1
- Setting up the mesh equation M_2 without the expression U_1

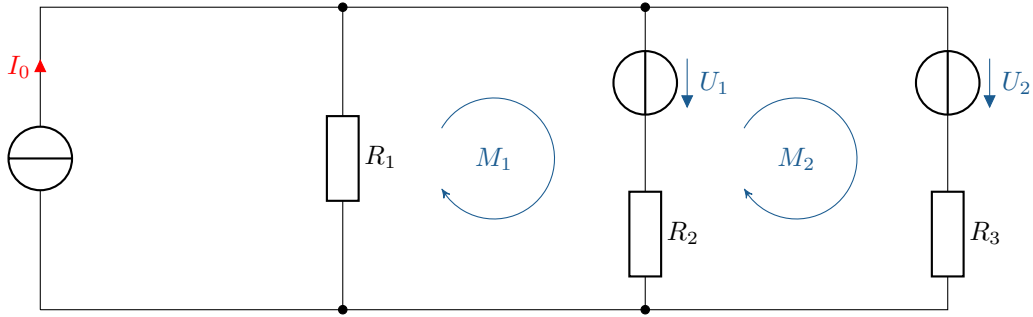
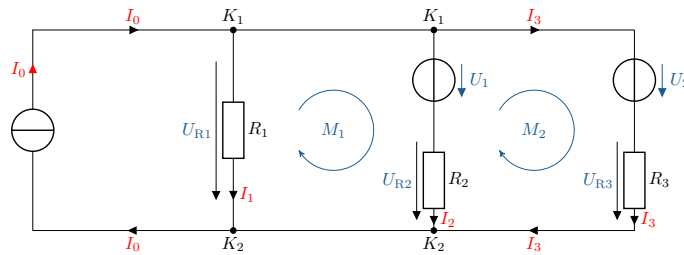


Figure 2.13: **Example.** Network analysis using Kirchhoff's laws.

- Specification of currents and voltages:



- Knot equations:

$$\begin{aligned} K_1 : I_0 - I_1 - I_2 - I_3 &= 0 \\ K_2 : -I_0 + I_1 + I_2 + I_3 &= 0 \end{aligned}$$

- Setting up the mesh equation M_1 and M_2 :

$$\begin{aligned} M_1 : U_1 + U_{R2} - U_{R1} &= 0 \\ M_2 : U_2 + U_{R3} - U_{R2} - U_1 &= 0 \end{aligned}$$

- Setting up the mesh equation M_2 without the expression U_1

$$\begin{aligned} M_1 : U_1 &= U_{R1} - U_{R2} \\ M_2 : U_2 + U_{R3} - (U_{R2} - (U_{R1} - U_{R2})) &= 0 \\ M_2 : U_2 + U_{R3} - \cancel{U_{R2}} - U_{R1} + \cancel{U_{R2}} &= 0 \\ M_2 : U_2 + U_{R3} - U_{R1} &= 0 \end{aligned}$$

3 Principle of superposition

After a basic analysis of meshes and nodes in electrical networks, increasingly complex electrical networks are considered. For this purpose, the superposition method according to Helmholtz is used. The superposition method is also referred to as the principle of superposition. In the superposition principle, all sources are evaluated individually one after the other and the results of the individual calculations are superimposed.



Learning objectives: Principle of superposition

The students

- understand the conditions of systems theory for the analysis of direct current networks.
- can apply the superposition principle to electrical networks.

3.1 Digression Systems theory

If only the input signal and the output signal are considered in a model, this is referred to as a black box. Here, no information is given about what happens between these two signals. Figure 3.1 describes such a model. Here, a black box is considered, which has an input variable F_E and an output variable F_A . In the black box, the input signal is transformed into the output signal. The image as a whole is referred to as a system.

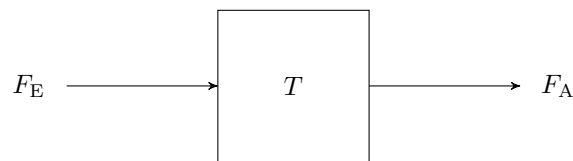


Figure 3.1: **Black box in systems theory.** A black box with input variables on the left and output variables on the right.

Based on the black box presented, the output signal F_A can be determined as a function of the input signal F_E . This transformation can be described by equation 3.1.

$$F_A = T(F_E) \quad (3.1)$$

An electrical network can also be considered such a system. The electrical network shown in Figure 3.2 has an input voltage U_E as the input variable and an output voltage U_A as the output variable. The black box, which represents the system to be transformed, consists of the two resistors R_1 and R_2 .

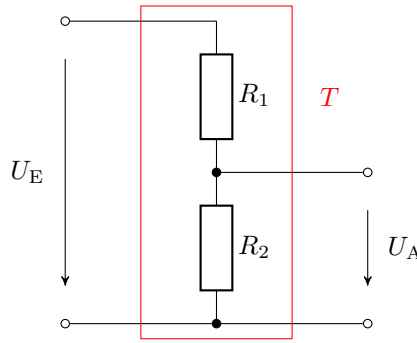


Figure 3.2: **Voltage divider as a black box.** Consideration of an electrical network as a black box with input voltage and output voltage.

The relationship between the output signal of a system and the input signal from Equation 3.1 still applies. The output signal can be described as a transformation of the input signal as in Equation 3.2. The output signal is equal to the voltage drop across resistor R_2 . This voltage can be calculated from the product of the input voltage and the resistance ratio. This relationship was also discussed in the section on voltage dividers in Chapter 4.

$$U_A = T(U_E) = U_E \cdot \frac{R_2}{R_1 + R_2} \quad (3.2)$$

Causality

If an output variable is based exclusively on the transformation of an input variable, there is a direct relationship between cause and effect. This also means that the output variable does not provide a changing system response prior to an excitation. A system in which these conditions prevail is referred to as a causal system (see Figure 3.3). For example, a DC excitation of a system produces a time-indifferent DC response. A system that behaves differently is referred to as a non-causal system. If the value of the input signal is zero for $t < t_0$, the value of the output signal must also be zero for the same period of time in order to fulfil the conditions of a causal system.

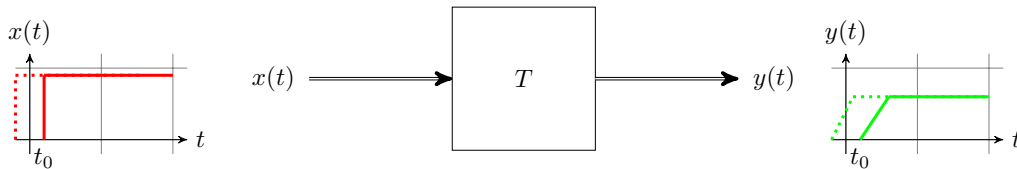


Figure 3.3: **Transformation of a causal system.** System transformed from $x(t)$ to $y(t)$ for a causal system.

Time variance

If the output signal of a system always responds to an input signal in the same way over time, this is referred to as a time-invariant system. If the timing of the input signal shifts, for example by t_0 , the output signal must still respond in the same way, based on the input signal, in time-invariant systems.

The stability of a system also describes a relationship between the input signal and the output signal. If an input signal with finite amplitude produces an output signal that does not grow beyond all limits, the system is stable. If the output signal continues to grow after the input signal has been deactivated, the system is unstable.

$$T(\alpha \cdot u_e) = \alpha \cdot T(u_e) \quad (3.3)$$

Linearity

If the response of the output signal is proportional to the excitation of the input signal, the system is described as a **linear** system (Equation 3.3). If two superimposed input signals act on a system, they can be considered separately and added together in this case. In this way, the transformations of the input signals, as in Equation 3.4, can also be considered separately.

$$T(u_1 + u_2) = T(u_1) + T(u_2) \quad (3.4)$$

Systems that are both time-invariant and linear are referred to as LTI (Linear Time Invariant) systems. The systems in this module are approximated as LTI systems. Components that exhibit non-linearities are dealt with, for example, in the chapter on periodic quantities.

Key point: LTI-Systems

LTI systems represent linear and time-invariant systems. DC networks must be considered as LTI systems so that the superposition theorem can be applied to them.

3.2 Superposition theorem

Once the electrical networks have been checked for linearity as the systems under consideration, the superposition theorem can be applied. Here, different sources can be considered separately from one another. In this way, the superposition theorem switches off all sources and switches on the individual sources in series. When sources are switched off, current sources and voltage sources are converted differently. When an ideal voltage source is switched off, a short circuit occurs at the location of the voltage source, as shown in Figure 3.4.

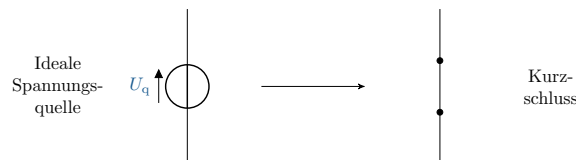


Figure 3.4: **Conversion of a voltage source.** If a voltage source is deactivated, a short circuit remains. This short circuit is used for further network calculations.

When switched off, a power source leaves only open terminals (Figure 3.5). This idle state prevents further consideration of this path in the network.

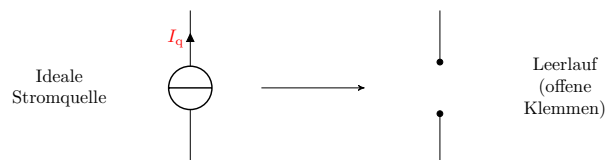


Figure 3.5: **Conversion of a power source.** Deactivating a power source leaves terminals open.

Merke: Conversion of sources

When voltage sources and current sources are switched off, deactivated voltage sources leave a short circuit and deactivated current sources leave an open circuit.

If all sources except one are switched off, the electrical network for the remaining source is analysed. This is then carried out sequentially with each source. At the end, the individual effects are considered as a sum. For example, the current I_R through a resistor R , which is supplied by two sources Q_1 and Q_2 , can be explained by the sum of the partial currents of the two sources (see Equation 3.5).

$$I_R = f(Q_1, Q_2) \quad (3.5)$$

To illustrate the superposition theorem, Figure 3.6 shows an electrical network with two voltage sources and three resistors. To calculate this network with more than one source, the network is considered separately for both voltage sources.

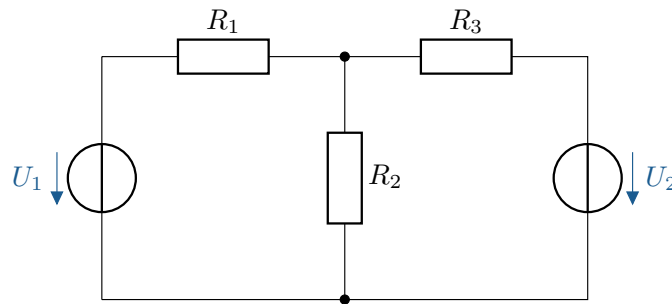


Figure 3.6: **Electrical network with two voltage sources to explain the superposition theorem.** The voltage sources should be analysed one after the other for the electrical network.

The electrical network presented is shown again in Figure 3.7 for the calculation of the two voltage sources separately. For the network analysis with voltage source U_1 , voltage source U_2 is short-circuited. The two resistors R_2 and R_3 are now connected in parallel with each other. The voltage of U_1 is now distributed across R_1 and the parallel connection R_{23} . Equivalently, when considering the network for the voltage source U_2 , the voltage source U_1 is short-circuited. Now the two resistors R_1 and R_2 form a parallel connection. The voltage of the voltage source U_2 is distributed across the parallel connection R_{12} and the resistor R_3 .

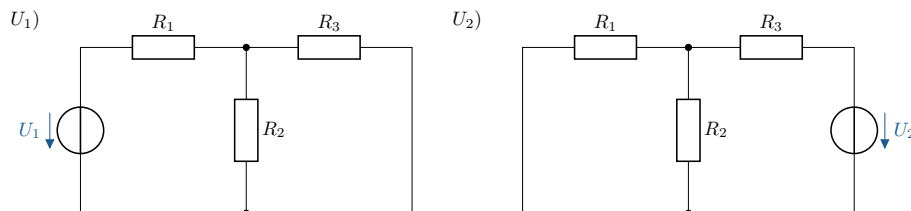


Figure 3.7: **Electrical network with two voltage sources to explain the superposition theorem.** On the left, the network for analysing the voltage source U_1 is shown, and on the right, the network for analysing the voltage source U_2 .

In equation 3.6 and equation 3.7, the voltages for the two voltage sources U_1 and U_2 at resistor R_3 are determined separately. In equation 3.8, the two voltages across the resistor R_3 are then added together according to the superposition theorem.

$$U_{R3}(U_1) = U_1 \cdot \frac{R_{23}}{R_1 + R_{23}} \quad (3.6)$$

$$U_{R3}(U_2) = U_2 \cdot \frac{R_{12}}{R_3 + R_{12}} \quad (3.7)$$

$$U_{R3} = U_{R3}(U_1) + U_{R3}(U_2) \quad (3.8)$$

**Merke: Überlagerungssatz**

Linear and time-invariant electrical networks with more than one source can be determined as the sum of the partial analyses of each individual source.

Beispiel 3.1: Superposition method

The electrical network shown in Figure 3.8 is given. The following tasks are to be completed:

Application of the superposition principle (Superposition theorem):

- Plotting currents and voltages
- Network calculation I_3 for U_g
- Network calculation I_3 for I_g
- How big is the current? I_3

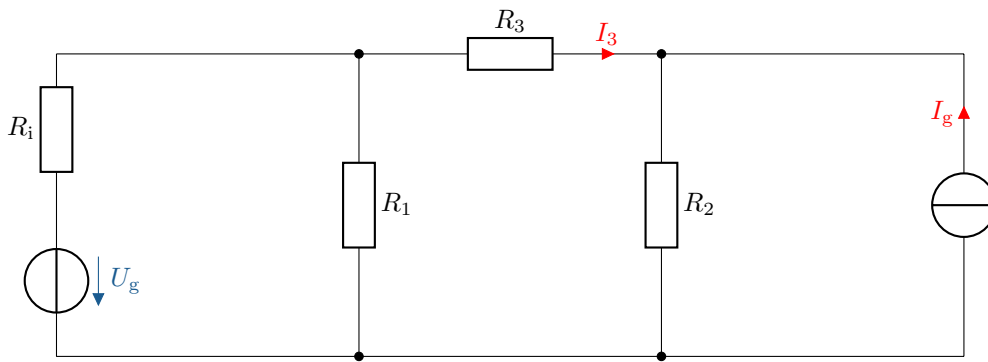
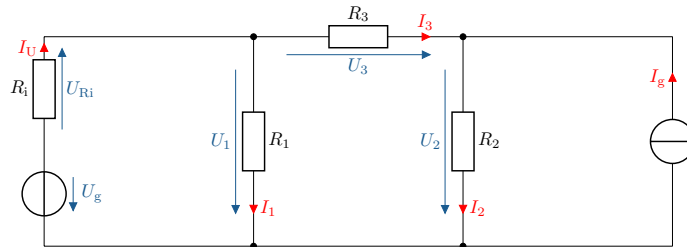
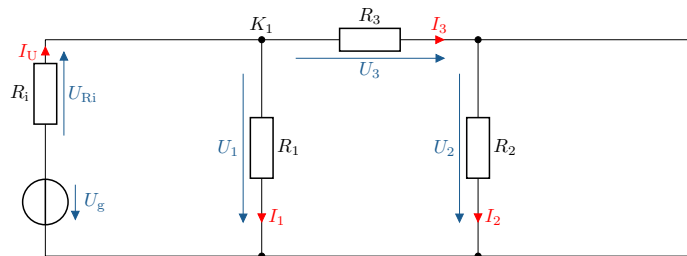


Figure 3.8: **Example.** Network analysis using Kirchhoff's rules.

- Specification of currents and voltages:



- Network calculation $I_3(U_g)$ for U_g :



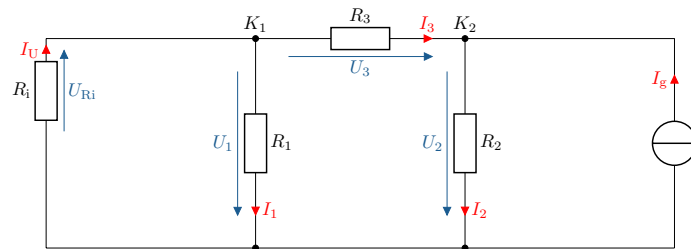
$$K_1 : I_{\text{ges}} = I_1 + I_3$$

$$I_{\text{ges}} = \frac{U_g}{R_{\text{ges}}} = \frac{U_g}{R_i + \frac{R_1 \cdot (R_2 + R_3)}{R_1 + R_2 + R_3}}$$

$$\frac{I_3}{I_{\text{ges}}} = \frac{R_1}{R_1 + R_2 + R_3}$$

$$I_3(U_g) = I_{\text{ges}} \cdot \frac{R_1}{R_1 + R_2 + R_3} = \frac{U_g}{R_i + \frac{R_1 \cdot (R_2 + R_3)}{R_1 + R_2 + R_3}} \cdot \frac{R_1}{R_1 + R_2 + R_3}$$

c) Network calculation $I_3(I_g)$ for I_g :



$$K_2 : I_{\text{ges}} = I_g = I_2 - I_3$$

$$\frac{-I_3}{I_{\text{ges}}} = \frac{R_2}{(R_1 || R_i) + R_2 + R_3}$$

$$I_3(I_g) = -I_g \cdot \frac{R_2}{(R_1 || R_i) + R_2 + R_3}$$

d) How strong is the current $I_3(U_g, I_g)$?

Superposition:

$$I_3(U_g, I_g) = I_3(U_g) + I_3(I_g)$$

$$I_3(U_g, I_g) = \frac{U_g}{R_i + \frac{R_1 \cdot (R_2 + R_3)}{R_1 + R_2 + R_3}} \cdot \frac{R_1}{R_1 + R_2 + R_3} + (-I_g \cdot \frac{R_2}{(R_1 || R_i) + R_2 + R_3})$$

4 Nodal analysis

Analysing an electrical network can sometimes be a complex task. As the network grows, so does the complexity of the analysis. In some cases, the analysis methods presented so far for node and mesh analysis are not sufficient to determine all the variables of an electrical network. In this case, the node potential method offers a possible solution for analysis. In the nodal analysis, all other potentials are determined based on a reference potential, with an assignment of 0V. Using the node potential method, the electrical network from Figure 4.1 is analysed as follows.

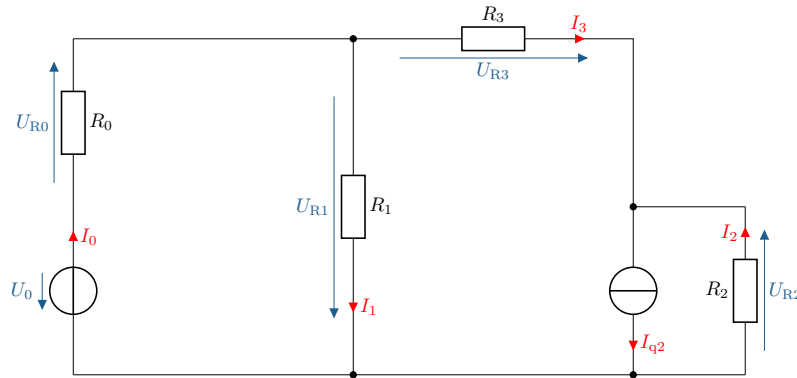


Figure 4.1: **Network for Nodal analysis.** Electrical network with a voltage source, a current source and four resistors. The network is used to explain the Nodal analysis.



Learning objectives: Node analysis

The Students

- are familiar with the steps of the nodal analysis
- can analyse electrical networks using the nodal analysis method.

4.1 Preparation of the network

The nodal analysis method works with current sources and conductance values. In addition to the current source, the network shown has a voltage source and resistance values. These need to be converted. As shown in Figure 4.2, the voltage source is transformed into a current source. The series resistance of the voltage source becomes a parallel resistance for the current source.

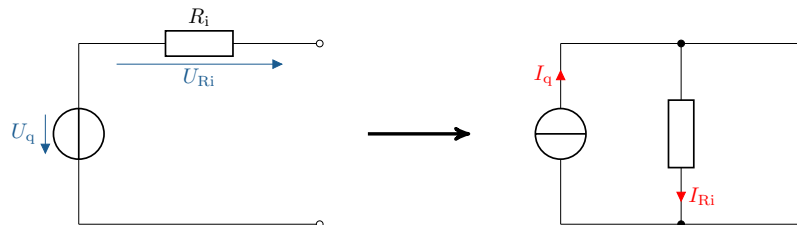


Figure 4.2: **Conversion of a voltage source.** Conversion of a voltage source with series resistance to a current source with parallel resistance.

When converting voltage sources into current sources, the voltage values of the voltage sources must

also be converted into current values. This is done according to equation 4.1.

$$I_q = U_q \cdot G_i \quad (4.1)$$

Since the nodal analysis method does not use the resistance values of the components, but rather the conductance values, all resistance values are converted into conductance values according to equation 4.2.

$$G_i = \frac{1}{R_i} \quad (4.2)$$

Figure 4.3 shows the conversion of the voltage sources from the specified network into equivalent current sources. The voltage source U_0 and the resistor R_0 become the current source I_{q0} and the parallel conductance G_0 . The remaining resistors are also converted into conductances.

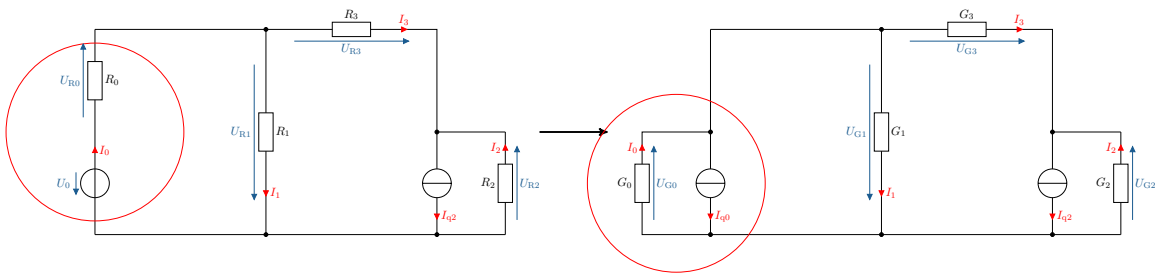


Figure 4.3: **Conversion of the network for the nodal analysis method.** The voltage source U_0 becomes the current source I_{q0} , and the resistance values R become conductances G .



Key point: Nodal analysis

In the nodal analysis method, an electrical network with current sources and conductances is analysed.

4.2 Determination of nodes and node potentials

Now the nodes must be determined. To do this, we define a reference node with the index 0, i.e. K_0 . Next, all nodes are numbered consecutively. The electrical network with the numbered nodes is shown in Figure 4.4. The nodes drawn without components between the numbered nodes have the same potential and thus form a node.

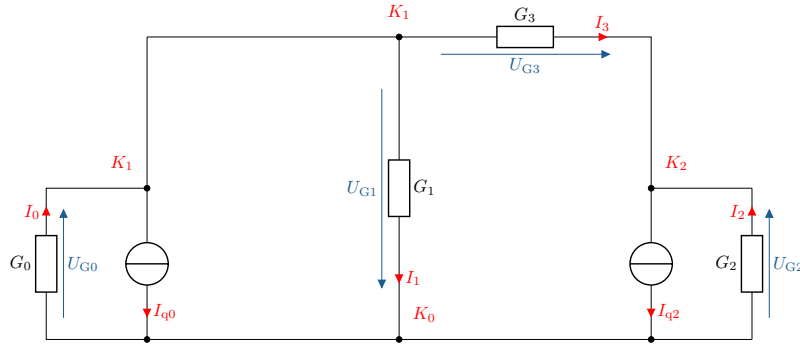


Figure 4.4: **Converted network.** Determination of the reference node K_0 and the consecutive nodes K_1 and K_2 .

To determine the node potentials, all potentials of nodes K_1 and K_2 are referenced to the reference node. $k - 1$ equations are required to perform the node potential method. Referring to the specified network, this results in two equations, which are written in vector notation according to equation 4.3. Here, the potential for node K_1 is defined by the voltage from node K_1 to the reference node. This results in the notation U_{K1-K0} . The same is done for node K_2 . Here, the voltage U_{K2-K0} is defined.

$$U_K = \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \end{bmatrix} \quad (4.3)$$

4.3 Allocation of source currents

For the specified nodes, apart from the reference node, the outgoing and incoming currents must be recorded. Each current source at the nodes is noted. Here, incoming current sources are given a positive sign and outgoing current sources are given a negative sign. The current from the current source $-I_{q0}$ flows out of node K_1 , and this current is assigned a negative sign. The current from the current source $-I_{q2}$ flows out of node K_2 , and this is also assigned a negative sign. According to equation 4.4, this results in the vector for the source currents.

$$I_K = \begin{bmatrix} -I_{q0} \\ -I_{q2} \end{bmatrix} \quad (4.4)$$

4.4 Conductance matrix

After converting the resistance values into conductance values, the conductance matrix is created using the conductance values. To do this, the nodes apart from the reference node are examined again. For this purpose, the electrical network presented is shown once more in Figure 4.5 with coloured branches. The conductance values of the nodes are noted along the main diagonal. For each node, the conductance values of the directly adjacent components are written down. For node K_1 , these would be the conductance values G_0 , G_1 and G_3 adjacent to the branch highlighted in blue, and for node K_2 , the conductance values G_2 and G_3 . Away from the main diagonal, the conductance values that lie directly between the nodes are recorded on the other elements of the conductance matrix. Between nodes K_1 and K_2 , the branch highlighted in green with the conductance value G_3 lies directly between them. If there are other nodes between the nodes under consideration, these are not directly connected to each other and a 0 is entered in the element of the conductance matrix. The conductance matrix created in this way is illustrated in equation 4.5.

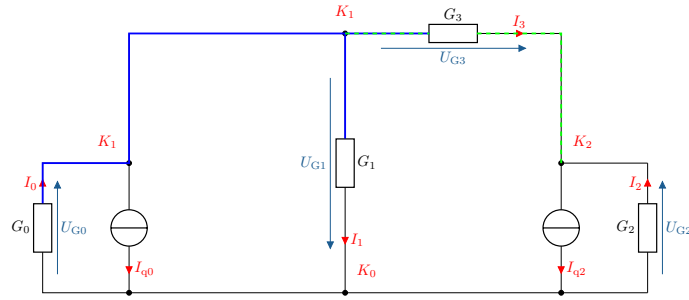


Figure 4.5: **Electrical network with highlighted connections.** Creation of the conductance matrix for the nodal analysis method

$$G_K = \begin{bmatrix} G_0 + G_1 + G_3 & -G_3 \\ -G_3 & G_2 + G_3 \end{bmatrix} \quad (4.5)$$

4.5 Set up a system of equations

After the vector of node potentials, the vector of source currents, and the conductance matrix have been determined, these equations are used to set up the system of equations for the node potential method, which is shown in Equation 4.5.

$$\begin{bmatrix} G_0 + G_1 + G_3 & -G_3 \\ -G_3 & G_2 + G_3 \end{bmatrix} \cdot \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \end{bmatrix} = \begin{bmatrix} -I_{q0} \\ -I_{q2} \end{bmatrix} \quad (4.6)$$

The system of equations of the node potential method can then be solved using the Gauss elimination method as an example. By rearranging the equation according to the node potentials sought, these can be identified. If, for example, the voltage across resistor R_1 has been calculated in the network under investigation, the current through the resistor can be determined. As an alternative to the Gaussian elimination method, Cramér's rule can also be used to solve linear systems of equations.

Beispiel 4.1: Knotenpotentialverfahren

Analysis of the electrical network from Figure 4.6 using the node potential method.

The following steps should be carried out for the node potential method:

- Preparation of the network
- Determination of nodes and node potentials
- Allocation of source currents
- Preparation of the conductivity matrix
- Set up a system of equations

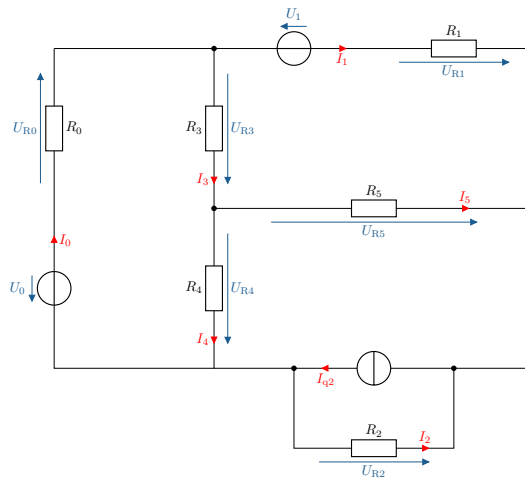
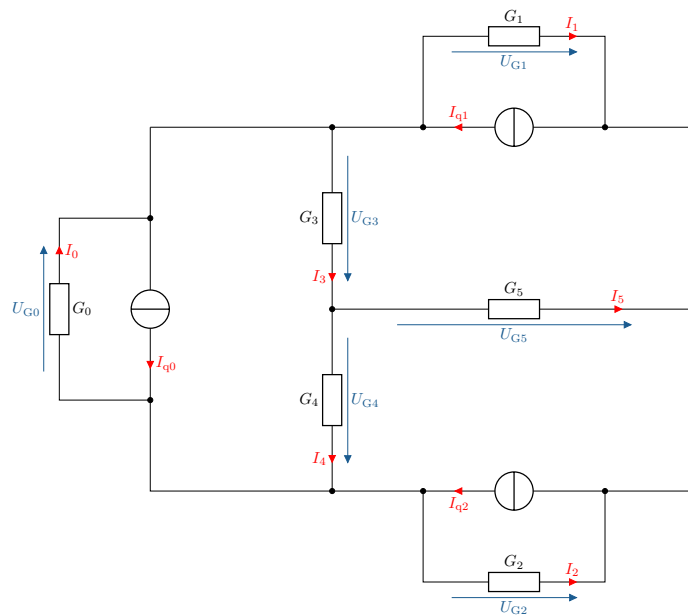
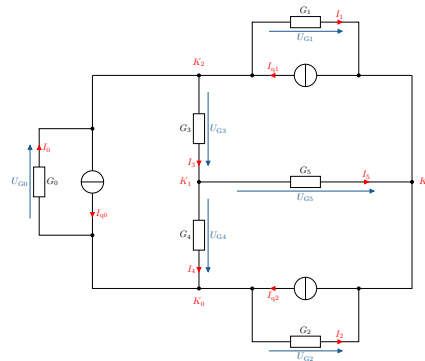


Figure 4.6: **Beispiel.** Node potential method on an electrical network.

- Preparing the network:



b) Determination of nodes and node potentials:



$$U_K = \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \\ U_{K3-K0} \end{bmatrix}$$

c) Allocation of source currents:

$$I_K = \begin{bmatrix} 0 \\ -I_{q0} + I_{q1} \\ -I_{q1} - I_{q2} \end{bmatrix}$$

d) Determination of the conductance matrix:

$$G_K = \begin{bmatrix} G_3 + G_4 + G_5 & -G_3 & -G_5 \\ -G_3 & G_0 + G_1 + G_3 & -G_1 \\ -G_5 & -G_1 & G_1 + G_2 + G_5 \end{bmatrix}$$

e) Set up a system of equations:

$$\begin{bmatrix} G_3 + G_4 + G_5 & -G_3 & -G_5 \\ -G_3 & G_0 + G_1 + G_3 & -G_1 \\ -G_5 & -G_1 & G_1 + G_2 + G_5 \end{bmatrix} \cdot \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \\ U_{K3-K0} \end{bmatrix} = \begin{bmatrix} 0 \\ -I_{q0} + I_{q1} \\ -I_{q1} - I_{q2} \end{bmatrix}$$

Key point: mesh flow method

The Students

- are familiar with the steps of the mesh flow process.
- can analyse electrical networks using the mesh current method.

5 Mesh flow method

In addition to node potential analysis for investigating electrical networks, the mesh current method can also be used for calculation. In the mesh current method, a system of equations is set up with which the mesh currents can be determined. In order to set up the system of equations for the mesh current method, some preparations must be made. In the following steps, the electrical network from Figure 5.1 is to be calculated using the mesh current method.

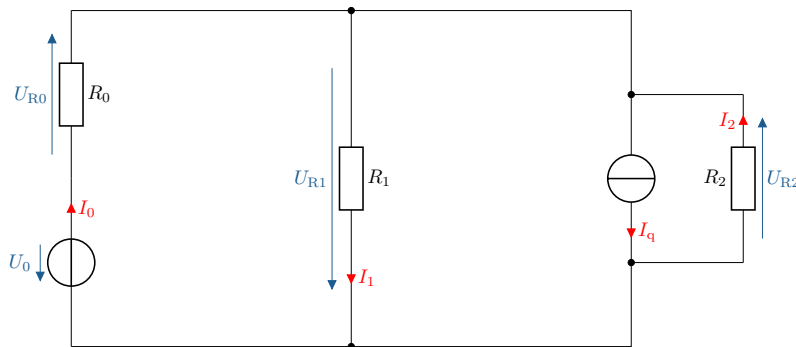


Figure 5.1: **Network for the mesh flow method.** Electrical network with a voltage source, a current source and three resistors. The network is used to explain the mesh current method..

5.1 Preparation of the network

The mesh current method works with voltage sources and resistance values. For this purpose, any current sources must be transformed into voltage sources. This type of conversion of a current source into a voltage source is shown once again in Figure 5.2. Here, the current source and the parallel internal resistance are transformed into a voltage source with a series internal resistance. Here, the current source and the parallel internal resistance are converted into a voltage source with a series internal resistance.

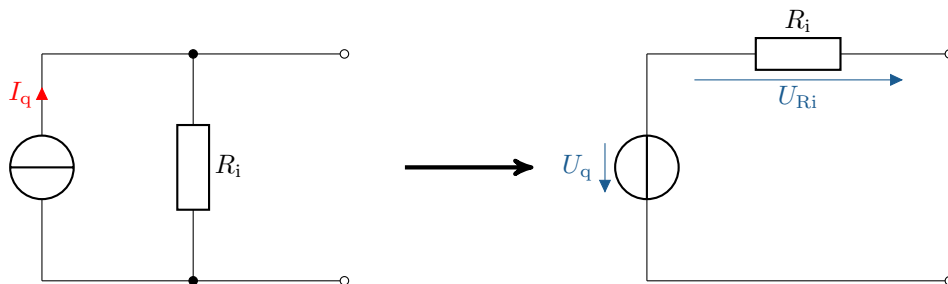


Figure 5.2: **Conversion of a power source.** Conversion of a current source with parallel resistance to a voltage source with series resistance.

Equation 5.1 shows the calculation of the voltage value of the voltage source from the specified current value of the current source and the resistance value of the internal resistance.

$$U_q = R_i \cdot I_q \quad (5.1)$$

In addition, possible conductance values of electrical resistors must be converted into resistance values. Equation 5.2 shows how the resistance value is determined from the reciprocal of the conductance.

$$R_i = \frac{1}{G_i} \quad (5.2)$$

In Figure 5.3, the current source of the network shown is converted into a voltage source. The parallel connection of the current source and resistor R_2 becomes a series connection consisting of the new voltage source and resistor R_2 .

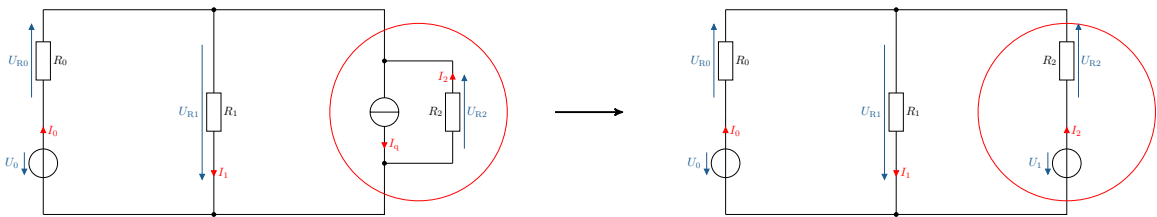


Figure 5.3: **Conversion of a power source in an electrical network.** Conversion of the current source with parallel resistance of the network to be calculated into a voltage source with series resistance. The transformation of the source takes place in the red circle.

Key point: Mesh flow method

For the mesh current method, voltage sources and resistance values are required to analyse an electrical network.

5.2 Define meshes

For mesh current analysis, the meshes must be defined. The mesh currents are determined via the meshes. In addition, the resistances are assigned to the meshes. Figure 5.4 shows the network presented with the meshes M_1 and M_2 marked.

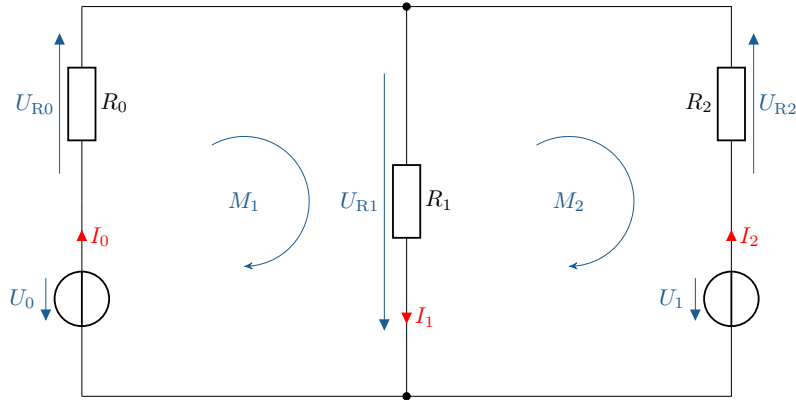


Figure 5.4: **Mesh in the electrical network.** Determination of meshes M_1 and M_2 and the associated mesh flows.

The mesh current I_{M1} flows through all components located at mesh M_1 . Similarly, all components at mesh M_2 are traversed by the mesh current I_{M2} . The two mesh currents I_{M1} and I_{M2} are noted vectorially in equation 5.3.

$$I_M = \begin{bmatrix} I_{M1} \\ I_{M2} \end{bmatrix} \quad (5.3)$$

5.3 Determine resistance matrix

In mesh M_1 , the mesh current I_{M1} encounters the two resistors R_0 and R_1 . Mesh M_1 is shown in blue in Figure 5.5. The mesh current I_{M2} of mesh M_2 flows through the resistors R_1 and R_2 . Resistor R_1 is therefore part of both meshes M_1 and M_2 and is traversed by both mesh currents. If a resistor is traversed by several mesh currents, it is considered a coupling resistor between the meshes. In the resistance matrix according to equation 5.4, the resistances of the meshes and the coupling resistances are assigned. Here the loop resistances from the meshes are entered in the main diagonal. This means that the series connection $R_0 + R_1$ becomes the element of the first column and row for the first mesh. The series connection $R_1 + R_2$ is entered in the element of the second column and row for the second mesh. The coupling resistances between the respective meshes are entered in the remaining elements of the resistance matrix. The coupling resistance is given a positive sign if both mesh currents flow in the same direction and a negative sign if they flow in opposite directions. In alternating current technology, the resistance matrix is also referred to as the mesh impedance matrix.

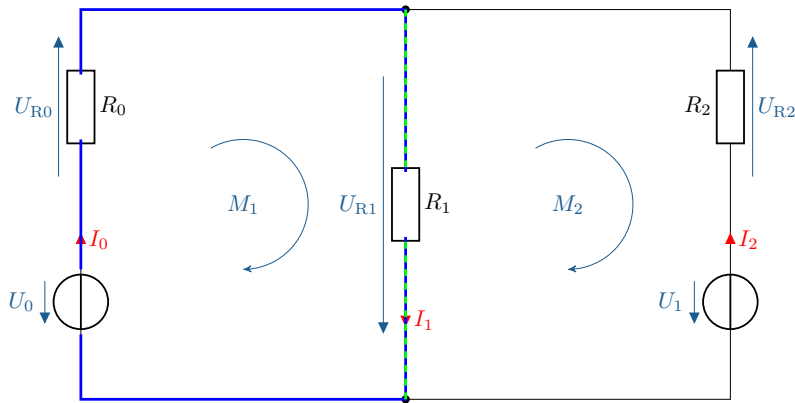


Figure 5.5: **Determination of the branches.** Electrical network with highlighted branches for creating the resistance matrix for the mesh current method

$$R_M = \begin{bmatrix} R_0 + R_1 & -R_1 \\ -R_1 & R_1 + R_2 \end{bmatrix} \quad (5.4)$$

5.4 Assign source voltages

The mesh must be assigned the stress sources, known as source stresses. The mesh M_1 has the source stress U_0 . The mesh M_2 has the source stress U_1 . If the direction of the voltage arrow and the mesh direction are the same, the source voltages have a negative sign. If the directions of the voltage arrow and the mesh circulation are opposite, the sign of the source voltage is positive. If there is no source in a mesh, a 0 is entered.

$$U_M = \begin{bmatrix} U_0 \\ -U_1 \end{bmatrix} \quad (5.5)$$

5.5 Set up a system of equations

According to Ohm's law, electrical voltage is calculated as the product of electrical resistance and current. In mesh current analysis, the resistance matrix R_M and the vector of source voltages U_M are used to calculate the vector of mesh currents I_M . The system of equations is represented in Equation 5.6.

$$\begin{bmatrix} R_0 + R_1 & -R_1 \\ -R_1 & R_1 + R_2 \end{bmatrix} \cdot \begin{bmatrix} I_{M1} \\ I_{M2} \end{bmatrix} = \begin{bmatrix} U_0 \\ -U_1 \end{bmatrix} \quad (5.6)$$

The quantities sought are calculated from the mesh currents. For example, the currents I_0 , I_1 and I_2 are sought. The results of the mesh currents are equivalent to the currents that flow through a single mesh current. In the electrical network shown, the current of mesh M_1 is $I_{M1} = I_0$ and the current of mesh M_2 is $I_{M2} = -I_2$. The current through resistor R_1 can be calculated using the node equation: $I_1 = I_0 + I_2$.

Beispiel 5.1: Mesh flow method

Analysis of the electrical network from Figure 5.6 using mesh current analysis.

The following steps should be carried out for mesh flow analysis:

- Preparation of the network
- Define meshes and mesh flows
- Determine resistance matrix
- Assign source voltages
- Set up a system of equations

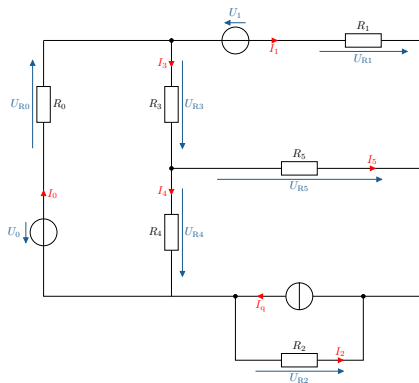
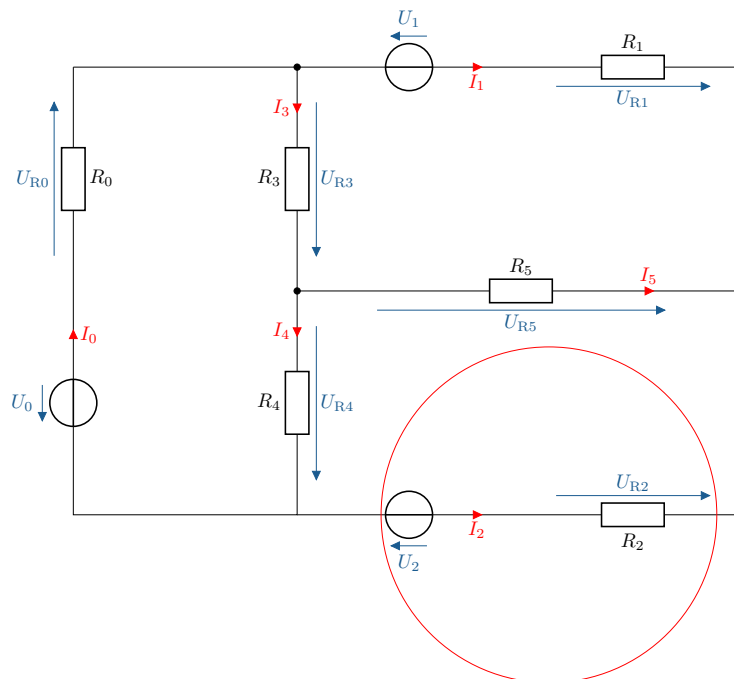
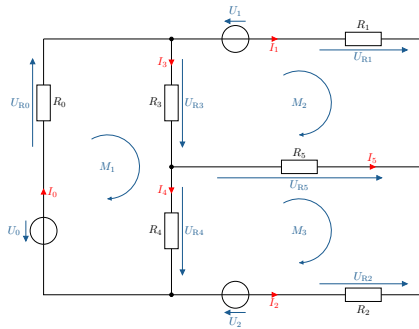


Abbildung 5.6: **Beispiel.** Maschenstromanalyse an einem elektrischen Netzwerk.

- Preparation of the network



- Define meshes and mesh flows



$$I_M = \begin{bmatrix} I_{M1} \\ I_{M2} \\ I_{M3} \end{bmatrix}$$

c) Determine resistance matrix

$$R_M = \begin{bmatrix} R_0 + R_3 + R_4 & -R_3 & -R_4 \\ -R_3 & R_1 + R_3 + R_5 & -R_5 \\ -R_4 & -R_5 & R_2 + R_4 + R_5 \end{bmatrix}$$

d) Assign source voltages

$$U_M = \begin{bmatrix} U_0 \\ U_1 \\ -U_2 \end{bmatrix}$$

e) Set up a system of equations

$$\begin{bmatrix} R_0 + R_3 + R_4 & -R_3 & -R_4 \\ -R_3 & R_1 + R_3 + R_5 & -R_5 \\ -R_4 & -R_5 & R_2 + R_4 + R_5 \end{bmatrix} \cdot \begin{bmatrix} I_{M1} \\ I_{M2} \\ I_{M3} \end{bmatrix} = \begin{bmatrix} U_0 \\ U_1 \\ -U_2 \end{bmatrix}$$

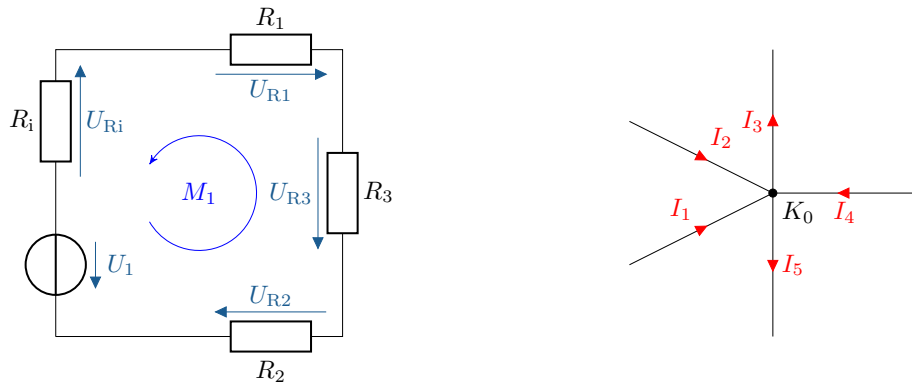
Calculate currents:

$$\begin{aligned} I_3 &= I_0 - I_1 \\ I_4 &= I_3 - I_5 \\ I_0 &= I_4 - I_2 \\ -I_5 &= I_2 + I_1 \end{aligned}$$

A Exercise tasks

A.1 Knots and meshes analysis 1

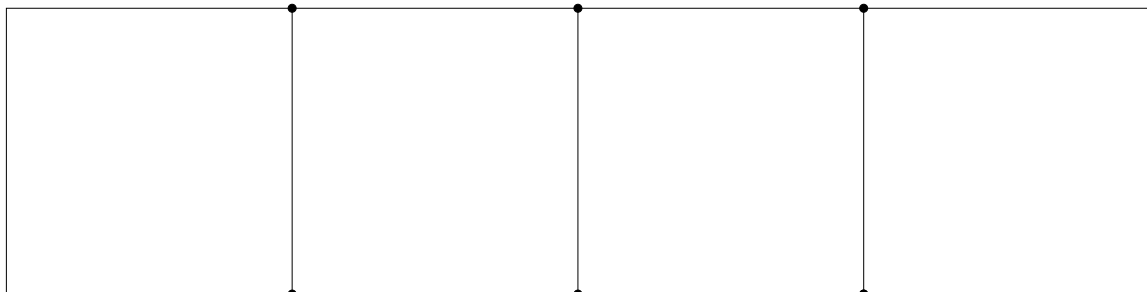
The following network is given below.



- Set up the mesh equation for mesh M_1 !
- Set up the knot equation for the knot K_0 !

A.2 Knots and meshes analysis 1

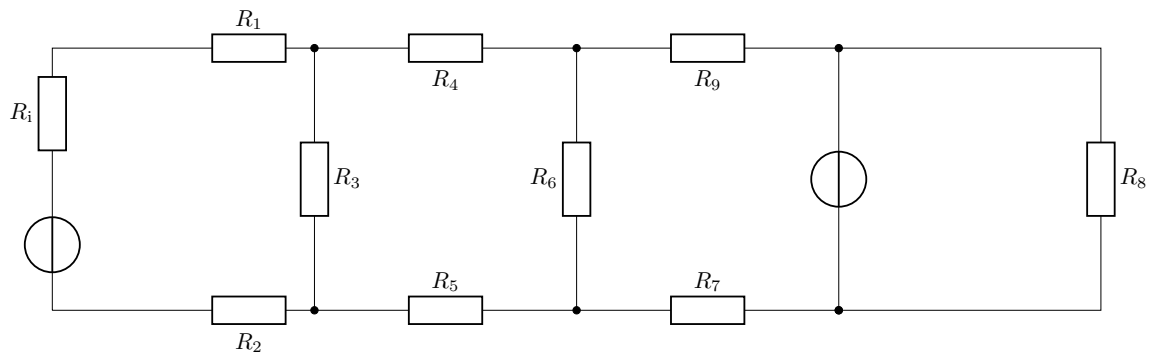
The following network is given below.



- How many knots, branches and meshes are needed for analysis?
- Draw a complete tree for the given network.

A.3 Knots and meshes analysis 3

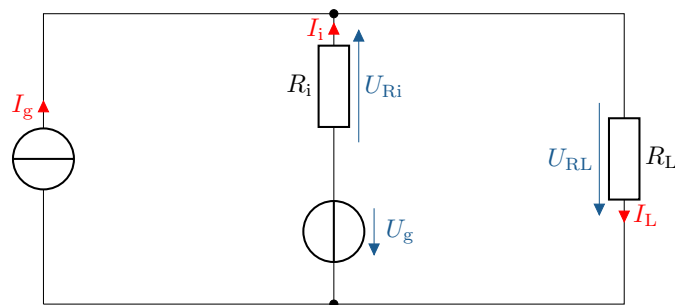
The following network is given below.



- Mark and label all currents and voltages!
- Set up the equations for all nodes!
- Set up the equations for all meshes!

A.4 Superposition 1

The following network is given below.

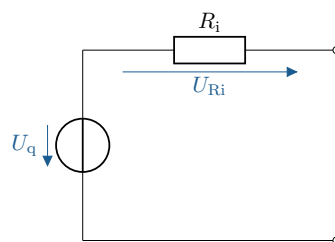


Using the superposition method, determine the voltage drop across the resistor R_L . To do this, follow these steps:

- Determine the proportion of U_g .
- Determine the proportion of I_g .
- Combine the results using the superposition theorem.

A.5 Nodal analysis method 1

Given is the network of a real voltage source $U_q = 3.6 \text{ V}$ with an internal resistance $R_i = 0.2 \Omega$.

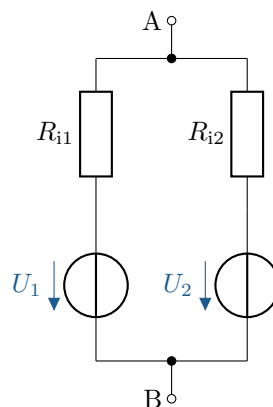


Convert the specified real voltage source into a real current source. To do this, follow these steps:

- a) Determine the conductance of the resistance value.
- b) Convert the real voltage source into a real current source.
- c) Calculate the current value of the converted current source.

A.6 Nodal analysis method 2

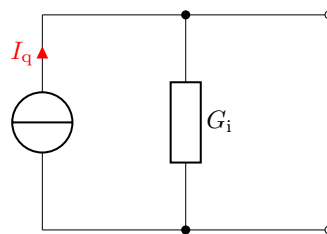
The following network shows two parallel equivalent circuits of battery cells. They consist of voltage sources U_1 and U_2 as well as internal resistances R_{i1} and R_{i2} . Perform the analysis using the node potential method.



- a) Rearrange the network so that the node potential method can be applied.
- b) Set up the bus admittance matrix.
- c) Set up the vector of the node inflows (I).
- d) Solve the system of equations and calculate the voltage U_0 between terminals A and B .

A.7 Mesh analysis 1

Given is the network of a real current source $I_q = 6\text{ A}$ with a conductance $G_i = \frac{2}{3}\text{ S}$.



Convert the specified real current source into a real voltage source. To do this, follow these steps:

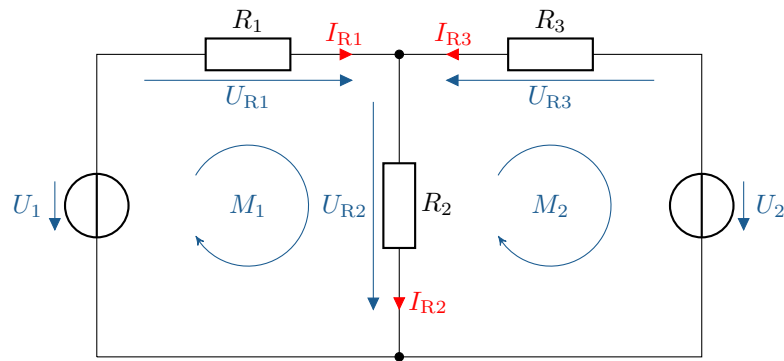
- a) Determine the resistance value of the conductance.
- b) Convert the real current source into a real voltage source.
- c) Calculate the voltage of the converted voltage source.

A.8 Mesh analysis 2

The following network shows a network consisting of three resistors and two voltage sources. Perform the analysis of the network using the mesh current method.

Werte:

$$R_1 = 4,7 \text{ k}\Omega \quad R_2 = 3,3 \text{ k}\Omega \quad R_3 = 2,2 \text{ k}\Omega \quad U_1 = 24 \text{ V} \quad U_2 = 12 \text{ V}$$



- Preparation of the network
- Define meshes and mesh flows
- Determine resistance matrix
- Assign source voltages
- Set up a system of equations

B Solutions to the exercises

B.1 Knots and meshes analysis 1

- Mesh equations:

$$M_1 : U_1 - U_{R1} - U_{R2} - U_{R3} = 0$$

- Knot equations:

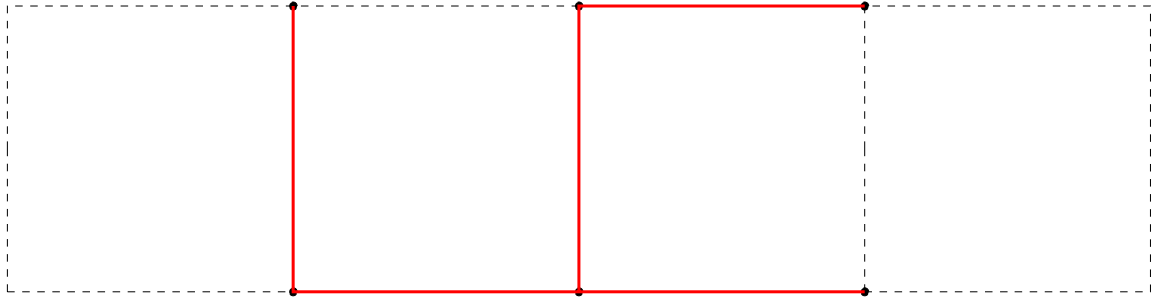
$$K_0 : +I_1 + I_2 - I_3 + I_4 - I_5 = 0$$

B.2 Knots and meshes analysis 1

- The network has:

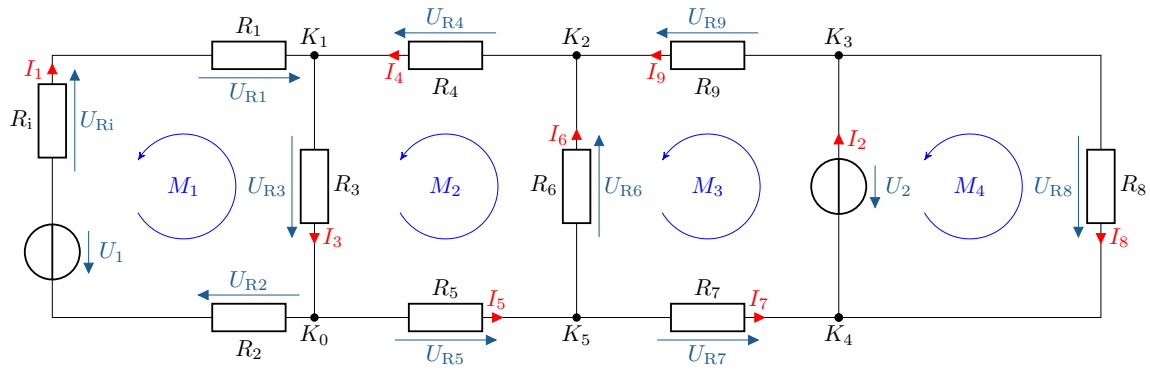
- Knot: $k=6$
- Branch: $k-1=5$
- Mesh: 4

b) Complete Tree



B.3 Knots and meshes analysis 3

a) Network with currents and voltages:



b) Knot equations:

$$K_0 : -I_1 + I_3 - I_5 = 0$$

$$K_1 : +I_1 - I_3 + I_4 = 0$$

$$K_2 : -I_4 + I_6 + I_9 = 0$$

$$K_3 : +I_2 - I_8 - I_9 = 0$$

$$K_4 : -I_2 - I_7 + I_8 = 0$$

$$K_5 : +I_5 - I_6 - I_7 = 0$$

c) Mesh equations

$$M_1 : U_1 - U_{Ri} - U_{R1} - U_{R2} - U_{R3} = 0$$

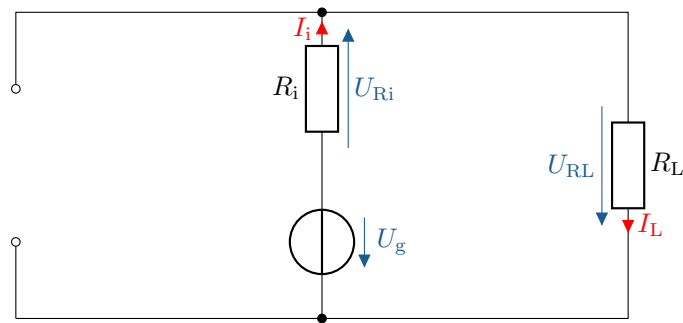
$$M_2 : U_{R3} + U_{R4} + U_{R5} + U_{R6} = 0$$

$$M_3 : -U_{R6} + U_{R7} - U_2 + U_{R9} = 0$$

$$M_4 : U_2 - U_{R8} = 0$$

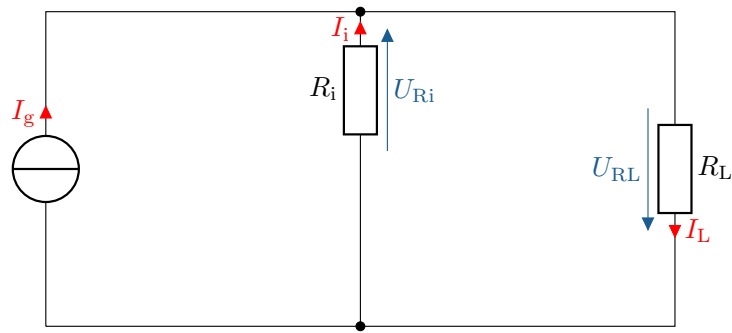
B.4 Superposition 1

a) Proportion of U_g :



$$U_{RL}(U_g) = \frac{R_L}{R_i + R_L} \cdot U_g$$

b) Proportion of I_g :



$$U_{RL}(I_g) = \frac{R_i \cdot R_L}{R_i + R_L} \cdot I_g$$

c) Superposition principle:

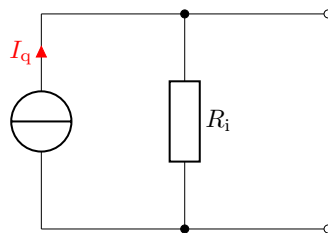
$$U_{RL} = U_{RL}(U_g) + U_{RL}(I_g) = \frac{R_L}{R_i + R_L} \cdot U_g + \frac{R_i \cdot R_L}{R_i + R_L} \cdot I_g$$

B.5 Nodal analysis method 1

a) Conductance:

$$G_i = \frac{1}{R_i} = \frac{1}{0,2 \, \Omega} = 5 \, \text{S}$$

b) Current source:

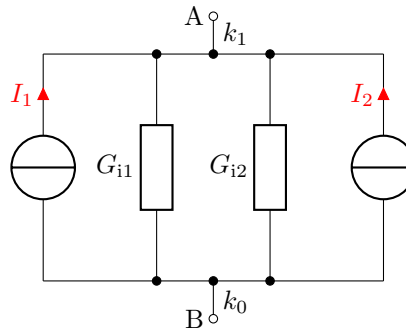


c) Current value

$$I_q = U_q \cdot G_i = 3,6 \text{ V} \cdot 5 \text{ S} = 72 \text{ A}$$

B.6 Nodal analysis method 2

a) Conversion:



$$I_1 = \frac{U_1}{R_{i1}} = \frac{0,8 \text{ V}}{4,8 \Omega} = 166 \text{ mA} \quad I_2 = \frac{U_2}{R_{i2}} = \frac{1,5 \text{ V}}{0,8 \Omega} = 1,875 \text{ A}$$

$$G_{i1} = \frac{1}{R_{i1}} = \frac{1}{4,8 \Omega} = 208 \text{ m}\Omega \quad G_{i2} = \frac{1}{R_{i2}} = \frac{1}{0,8 \Omega} = 1,25 \Omega$$

b) Bus admittance matrix:

$$KAM = (G_{i1} + G_{i2}) = 208 \text{ m}\Omega + 1,25 \Omega$$

c) Vector of node inflows:

$$I = (I_1 + I_2) = 166 \text{ mA} + 1,875 \text{ A}$$

d) Solve the system of equations:

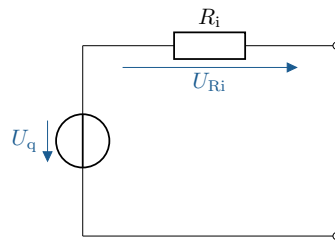
$$\begin{aligned} KAM \cdot U &= I \\ (G_{i1} + G_{i2}) \cdot U_0 &= (I_1 + I_2) \\ (208 \text{ m}\Omega + 1,25 \Omega) \cdot U_0 &= (166 \text{ mA} + 1,875 \text{ A}) \\ U_0 &= \frac{I_1 + I_2}{G_{i1} + G_{i2}} = \frac{166 \text{ mA} + 1,875 \text{ A}}{208 \text{ m}\Omega + 1,25 \Omega} = 1,4 \text{ V} \end{aligned}$$

B.7 Mesh analysis 1

a) Resistance value:

$$R_i = \frac{1}{G_i} = \frac{1}{\frac{2}{3} \text{ S}} = 1,5 \Omega$$

b) Voltage source:



c) Voltage value

$$U_q = R_i \cdot I_q = 1,5 \, \Omega \cdot 6 \, A = 9 \, V$$

B.8 Mesh analysis 2

a) Preparing the network:
The network consists of:

- Two Voltage sources: U_1 und U_2 ,
- Three resistors: R_1 , R_2 , R_3 ,
- Three possible stitches.

Voltage sources and resistance values are available.

b) Define meshes and mesh flows:

- M_1 : Mesh flow in the left mesh,
- M_2 : Mesh flow in the right mesh,
- The current I_{R2} divided by R_2 results in:

$$I_{R2} = \begin{bmatrix} I_{M1} \\ I_{M2} \end{bmatrix}$$

c) Determine resistance matrix:

$$\mathbf{R}_M = \begin{bmatrix} R_1 + R_2 & -R_2 \\ -R_2 & R_2 + R_3 \end{bmatrix}$$

Inserting the values:

$$\mathbf{R}_M = \begin{bmatrix} 4,7 \, \Omega + 3,3 \, \Omega & -3,3 \, \Omega \\ -3,3 \, \Omega & 3,3 \, \Omega + 2,2 \, \Omega \end{bmatrix} = \begin{bmatrix} 8,0 \, \Omega & -3,3 \, \Omega \\ -3,3 \, \Omega & 5,5 \, \Omega \end{bmatrix}$$

d) Assign source voltages:

$$\mathbf{U}_M = \begin{bmatrix} U_1 \\ -U_2 \end{bmatrix} = \begin{bmatrix} 24 \, V \\ -12 \, V \end{bmatrix}$$

e) Solve a system of equations:

$$\mathbf{R}_M \cdot \mathbf{I}_M = \mathbf{U}_M,$$

whereby:

$$\mathbf{R}_M = \begin{bmatrix} 8,0 \, \Omega & -3,3 \, \Omega \\ -3,3 \, \Omega & 5,5 \, \Omega \end{bmatrix}, \quad \mathbf{I}_M = \begin{bmatrix} I_{M1} \\ I_{M2} \end{bmatrix}, \quad \mathbf{U}_M = \begin{bmatrix} 24 \, V \\ -12 \, V \end{bmatrix}.$$

Solving the system of equations gives:

$$I_{M1} = 2,79 \, \text{mA} \quad \text{und} \quad I_{M2} = -0,507 \, \text{mA}$$

This results in:

$$I_{R1} = I_{M1} = 2,79 \, \text{mA}$$

$$I_{R2} = I_{M1} + (-I_{M2}) = 2,79 \, \text{mA} + 0,507 \, \text{mA} = 3,297 \, \text{mA}$$

$$I_{R3} = -I_{M2} = 0,507 \, \text{mA}$$

and

$$U_{R1} = R_1 \cdot I_{R1} = 4,7 \, k\Omega \cdot 2,79 \, \text{mA} = 13,12 \, V$$

$$U_{R2} = R_2 \cdot I_{R2} = 3,3 \, k\Omega \cdot 3,297 \, \text{mA} = 10,88 \, V$$

$$U_{R3} = R_3 \cdot I_{R3} = 2,2 \, k\Omega \cdot 0,507 \, \text{mA} = 1,12 \, V$$

Checking the mesh:

$$U_1 = U_{R1} + U_{R2} = 24 \, V$$

$$U_2 = U_{R2} + U_{R3} = 12 \, V$$